

13 Gravitational fields

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Gravitational field strength is force per unit mass, $g = GM/r^2$ outside a spherical mass. Potential is work per unit mass relative to infinity, $\phi = -GM/r$. The negative sign records an attractive bound state. Circular satellites satisfy $GMm/r^2 = mv^2/r$; a geostationary orbit has Earth's rotational period.

Worked example: follow the reasoning

If orbital radius doubles about the same mass, g becomes one quarter of its former value. Potential magnitude is halved because it scales as $1/r$.

Quick check · multiple choice

A How does gravitational field strength vary outside a spherical mass?

[1]

- A As $1/r^2$
- B As r^2
- C As $1/r$
- D Constant with r

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 At radius r , orbital speed is v . Derive the new speed for a circular orbit of radius $4r$ about the same central mass.

[2]

Final answer: _____

Answers and explanation

Quick check: A. As $1/r^2$

Question 1 [2 marks]

Equate $GMm/r^2 = mv^2/r$ to give $v = \sqrt{GM/r}$ [1]. At $4r$, speed is $v/2$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.